

Ricardo Andrés Calvo Méndez

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SUMMARY

Ph.D. student in Electrical and Computer Engineering at Purdue University working on **formal verification**, **program analysis**, **software security**, and **performance optimization**. My research automates program analysis for vulnerability discovery and optimization opportunities. I bring **3+ years of industry experience** in full-stack development and quality assurance, including leading development teams.

Research interests: formal verification; program analysis and testing; performance optimization.

EDUCATION

Ph.D. in Electrical and Computer Engineering Aug 2025 – Expected May 2030
Purdue University, West Lafayette, IN

B.Sc. in Systems and Computer Engineering Feb 2018 – Oct 2023
Universidad Nacional de Colombia (ranked #1 in Colombia), Bogotá, Colombia. Class rank: 2/49.

RESEARCH EXPERIENCE

Graduate Research Assistant Aug 2025 – Present
Purdue University — Advisor: James C. Davis

- Co-develop *AutoSOUP*, a system that combines large language models with the CBMC bounded model checker to automate memory-safety verification of C components (ACM CCS 2026); contributed to system design.
- Designed AutoSOUP's **debugging agent**, which reads verification errors, diagnoses why a proof fails, and proposes a fix.
- Built the experimental setup used to run AutoSOUP's large-scale evaluation on a computing cluster.
- Across **177 components** from four embedded operating systems, AutoSOUP exposed **66.7% of 60 known CVEs** (vs. 28.3% for a frontier coding agent) and uncovered **20 new vulnerabilities** reported to maintainers.
- **Co-first author** of a large-scale study of how AI coding agents optimize software, analyzing **2,260 performance pull requests** (1,130 agentic, 1,130 human); extends our MSR 2026 paper, which I presented.
- Co-authored a study on false-positive crashes in Google's OSS-Fuzz-Gen (FSE 2026); presenting author.

Undergraduate Visiting Scholar (UREP-C) Feb 2023 – Jul 2023
Purdue University — Advisor: James C. Davis

- With mentorship, designed and implemented the fuzzing strategy of *EmNetTest*, a packet-validation testing framework, and its evaluation on **4 embedded network stacks**, using C, C++ and Python.
- Contributed to the manual triage of results; EmNetTest reproduced **12 known vulnerabilities** and found **7 new defects (6 CVEs)**, leading to an ASE 2023 publication.

PROFESSIONAL EXPERIENCE

Test Engineer & QA Analyst Sep 2024 – Jul 2025
Kravata S.A.S., Bogotá, Colombia

- Joined as the company's **first and only QA engineer** and defined its testing strategy, standards, and processes from scratch.
- Led testing for the company's **4 most critical services**.
- Automated unit, integration and end-to-end tests with Python, Cucumber, Selenium and Flask, and integrated them into CI/CD pipelines.

Full-Stack Software Developer

Aug 2023 – Aug 2024

Fit Ideas S.A.S., Bogotá, Colombia

- Promoted to team lead; led **4 developers** building and deploying a **white-label** cross-platform fuel sales application, customizable per client brand.
- Defined the system architecture, ran code reviews, and supported the team's Scrum process.
- Built web and mobile systems with Django, React, Spring Boot, Flutter and PostgreSQL, including end-to-end tests.

Junior Software Developer

Aug 2021 – Jan 2023

Fit Ideas S.A.S., Bogotá, Colombia

- Built a configurable web scraper that collects public online data on demand.
- Developed web and mobile product features with Django, React, Spring Boot and Flutter, and designed and maintained PostgreSQL databases.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Fall 2026

ECE 30861: Software Engineering — Purdue University

- Grade assignments and hold office hours for a course of ~150 students.

TECHNICAL SKILLS

Research & Security: Formal verification, memory-safety verification, bounded model checking (CBMC), program analysis, fuzzing, performance optimization

Programming: C, C++, Python, Java, JavaScript, TypeScript, Dart, SQL

AI for SE: LLM agents, agentic pipelines, empirical evaluation

Engineering: Spring Boot, React, Django, Flask, Flutter, PostgreSQL, Selenium, Cucumber, CI/CD, Docker, Linux, Git

REFEREED CONFERENCE PUBLICATIONS

- [C-1] P.C. Amusuo, R. Calvo, D. Anandayavaraj, T. Le Lievre, K. Kolyakov, E. Jorgensen, A. Machiry, and J.C. Davis. *AutoSOUP: Safety-Oriented Unit Proof Generation for Component-level Memory-Safety Verification* *arXiv*. Accepted at the 33rd ACM Conference on Computer and Communications Security (CCS'26). *Presenting author*.
- [C-2] P. Amusuo, D. Liu, R. Calvo, J. Metzman, O. Chang, and J.C. Davis. *Lessons from Mitigating False Positives in Google's OSS-Fuzz-Gen* *DOI*. Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE-Industry'26). *Presenting author*.
- [C-3] H. Peng, A. Z. Qiu, R. Calvo, K.G. Kalu, and J.C. Davis. *How Do Agents Perform Code Optimization? An Empirical Study* *DOI*. Proceedings of the 23rd International Conference on Mining Software Repositories — Mining Challenge track (MSR-Mining'26). *Presenting author*.
- [C-4] P.C. Amusuo, R. Calvo, Z. Xu, A. Machiry, and J.C. Davis. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks* *DOI*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'23).

PRELIMINARY WORKS

- [P-1] H. Peng*, R. Calvo*, K.G. Kalu, and J.C. Davis. *How Do Coding Agents Optimize Software and Report Performance Validation? A Large-Scale Empirical Study of Open-Source Pull Requests* *arXiv*. Technical preprint (2026). **Equal contribution*.

[P-2] P.C. Amusuo, D. Liu, R. Calvo, J. Metzman, O. Chang, and J.C. Davis. *FalseCrashReducer: Mitigating False Positive Crashes in OSS-Fuzz-Gen Using Agentic AI* [arXiv](#). Technical preprint (2025).

ACADEMIC SERVICE

Reviewer — MSR'26 Mining Challenge Track	2026
Sub-reviewer — ICSE'27	2026
Sub-reviewer — ICSE'26	2026
Sub-reviewer — USENIX Security '26 (Cycle 2)	2026
Student Volunteer — ICSE'26	2026

GRANTS AND SCHOLARSHIPS

CCS'26 Student Conference Grant <i>ACM CCS</i> <i>Travel support for CCS in The Hague, Netherlands.</i>	2026
ICSE'26 Student Mentoring Workshop (SMeW) <i>National Science Foundation (NSF) and ACM SIGSOFT</i> <i>Travel support for the Student Mentoring Workshop in Rio de Janeiro, Brazil.</i>	2026
“Ser Pilo Paga” Scholarship <i>Ministry of Education of Colombia</i> <i>Merit scholarship for a 99th-percentile national exam score; full undergraduate tuition.</i>	2017